Inverse Trigonometric Function: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Meaning of ITF

KEY POINTS

- Inverse exists only for a bijection: restrict $\sin x$ to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\cos x$ to $[0, \pi]$, $\tan x$ to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\cot x$ to $(0, \pi)$
- $\sin^{-1} x = \arcsin x$ is an angle (principal value), never $\frac{1}{\sin x}$
- Graph of f^{-1} = mirror image of the restricted graph in $y = x$

2. Domain and range

Formulas

$$\sin^{-1} x : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \cos^{-1} x : [-1, 1] \to [0, \pi]$$

$$\tan^{-1} x : R \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \cot^{-1} x : R \to (0, \pi)$$

$$\sec^{-1} x : |x| \geq 1 \to [0, \pi] - \left\{\frac{\pi}{2}\right\}, \quad \operatorname{cosec}^{-1} x : |x| \geq 1 \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

KEY POINTS

- First family $\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}$: odd; second family $\cos^{-1}, \cot^{-1}, \sec^{-1}$: $f(x) + f(-x) = \pi$
- Domain of $\sin^{-1} u$: $-1 \leq u \leq 1$; of $\sec^{-1} u$: $u \leq -1$ or $u \geq 1$
- Range of $f^{-1}(u)$: find the set of u, then apply the monotonic f^{-1}

Watch out

$\tan^{-1} x$, $\cot^{-1} x$ never reach $\pm\frac{\pi}{2}$, 0, π (open ends)

3. Graph facts

KEY POINTS

- Increasing: $\sin^{-1}, \tan^{-1}$; decreasing: $\cos^{-1}, \cot^{-1}$
- All bounded and aperiodic; $x = 0$ is the point of inflection of the first four
- Concave up for $x > 0$: $\sin^{-1}, \cot^{-1}$; for $x < 0$: $\cos^{-1}, \tan^{-1}$
- Max/min: $\sin^{-1}$: $\pm\frac{\pi}{2}$ at $x = \pm 1$; $\cos^{-1}$: π at -1, 0 at 1

Formulas

$(\sin^{-1} x)' = \dfrac{1}{\sqrt{1 - x^2}}$, $(\tan^{-1} x)' = \dfrac{1}{1 + x^2}$, $(\sec^{-1} x)' = \dfrac{1}{|x|\sqrt{x^2 - 1}}$; co-functions: minus sign

4. Inequalities

Formulas

$$\sin^{-1} x > \sin^{-1} y \iff 1 \geq x > y \geq -1; \quad \tan^{-1} x > \tan^{-1} y \iff x > y$$

$$\cos^{-1} x > \cos^{-1} y \iff -1 \leq x < y \leq 1; \quad \cot^{-1} x > \cot^{-1} y \iff x < y$$

STEPS

1. Constant inside the range: apply sin/tan (sign kept) or cos/cot (sign reversed)
2. Constant outside the range: true for the whole domain or for no x
3. Quadratic in $t = \sin^{-1} x$: solve for t, keep t in the range, then convert
4. Intersect with the domain

5. Evaluation and property 1

STEPS

1. Put θ = the inverse; draw a right triangle; read the required ratio
2. Fix the sign from the quadrant of θ (principal range)

3. For 2θ, 3θ, $\frac{\theta}{2}$ use multiple / half-angle formulas

KEY POINTS

- $\sin(\sin^{-1} x) = x$ only for $|x| \le 1$; $\tan(\tan^{-1} x) = x$ for all x; $\sec(\sec^{-1} x) = x$ for $|x| \ge 1$

6. Property 2: $f^{-1}(f(x))$

Formulas

$\sin^{-1}(\sin x) = x - n\pi$ (n even), $n\pi - x$ (n odd); $n\pi$ nearest to x
$\cos^{-1}(\cos x) = |x - 2k\pi|$ ($2k\pi$ nearest to x)
$\tan^{-1}(\tan x) = x - n\pi$ ($n\pi$ nearest); $\cot^{-1}(\cot x) = x - \left[\frac{x}{\pi}\right]\pi$

KEY POINTS

- $\sin^{-1}(\sin x)$: zig-zag, period 2π, odd; $\cos^{-1}(\cos x)$: triangle wave, period 2π, even
- $\tan^{-1}(\tan x)$, $\cot^{-1}(\cot x)$: saw-tooth of slope 1, period π
- $\sec^{-1}(\sec x) = \cos^{-1}(\cos x)$ graph; $\text{cosec}^{-1}(\text{cosec} x) = \sin^{-1}(\sin x)$ graph, minus excluded points
- Number of solutions of $f^{-1}(f(x)) = g(x)$: count intersections of the two graphs

Watch out

For $f(kx)$ first find the interval of kx, then split at turning points

7. Trigonometric substitution

KEY POINTS

- $\sqrt{1-x^2}$: $x = \sin\theta$ or $\cos\theta$; $1 + x^2$: $x = \tan\theta$; $\frac{1-x}{1+x}$: $x = \cos 2\theta$ or $\tan\theta$
- After substitution apply property 2 to $f^{-1}(f(k\theta))$ according to the interval of $k\theta$

Formulas

$$\sin^{-1}\frac{2x}{1+x^2} = 2\tan^{-1} x \ (|x| \le 1),\ \pi - 2\tan^{-1} x \ (x > 1),\ -\pi - 2\tan^{-1} x \ (x < -1)$$

$$\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1} x \ (x \ge 0),\ -2\tan^{-1} x \ (x < 0)$$

$$\tan^{-1}\frac{2x}{1-x^2} = 2\tan^{-1} x \ (|x| < 1),\ \pi + 2\tan^{-1} x \ (x < -1),\ -\pi + 2\tan^{-1} x \ (x > 1)$$

$$\sin^{-1}(3x - 4x^3) = 3\sin^{-1} x \ \left(|x| \le \frac{1}{2}\right);\ \cos^{-1}(4x^3 - 3x) = 3\cos^{-1} x \ \left(\frac{1}{2} \le x \le 1\right)$$

8. Properties 3, 4, 5

Formulas

$$\sin^{-1}(-x) = -\sin^{-1} x,\ \tan^{-1}(-x) = -\tan^{-1} x;\ \cos^{-1}(-x) = \pi - \cos^{-1} x,\ \cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$\sin^{-1} x + \cos^{-1} x = \tan^{-1} x + \cot^{-1} x = \sec^{-1} x + \text{cosec}^{-1} x = \frac{\pi}{2}$$

$$\text{cosec}^{-1} x = \sin^{-1}\frac{1}{x},\ \sec^{-1} x = \cos^{-1}\frac{1}{x};\ \cot^{-1} x = \tan^{-1}\frac{1}{x}\ (x > 0),\ \pi + \tan^{-1}\frac{1}{x}\ (x < 0)$$

KEY POINTS

- Expressions in $\sin^{-1} x$ and $\cos^{-1} x$: put $t = \sin^{-1} x$, $\cos^{-1} x = \frac{\pi}{2} - t$, find the range in t
- $(\sin^{-1} x)^2 + (\cos^{-1} x)^2 \in \left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$; $(\tan^{-1} x)^3 + (\cot^{-1} x)^3 \in \left[\frac{\pi^3}{32}, \frac{7\pi^3}{8}\right)$

Watch out

$\cot^{-1} x = \tan^{-1}\frac{1}{x}$ is false for $x < 0$

9. Addition and subtraction

Formulas

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy}\ (xy < 1);\ \pi + \tan^{-1}\frac{x+y}{1-xy}\ (x, y > 0,\ xy > 1)$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy} \quad (xy > -1)$$

$$\sin^{-1}x + \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) \quad (x, y \geq 0,\ x^2+y^2 \leq 1)$$

$$\cos^{-1}x - \cos^{-1}y = \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right) \quad (x \leq y)$$

KEY POINTS

- Three angles: $\tan(A+B+C) = \frac{S_1 - S_3}{1 - S_2}$
- Factor $\frac{1}{2}$: $\tan\left(\frac{1}{2}\cos^{-1}x\right) = \sqrt{\frac{1-x}{1+x}}$, $\cos\left(\frac{1}{2}\cos^{-1}x\right) = \sqrt{\frac{1+x}{2}}$
- Convert $\sin^{-1}$, $\cos^{-1}$, $\cot^{-1}$ of positive numbers to $\tan^{-1}$ before adding

Watch out

Check $xy < 1$ or $xy > 1$ before using the formula; the answer may need $+\pi$

10. ITF equations

STEPS

1. Write domain conditions of all terms
2. Reduce to one ITF with property 4, or combine angles
3. Take sin/cos/tan of both sides; solve
4. Substitute back: reject roots that put a side outside its range

KEY POINTS

- Sum of ITFs equal to the sum of their maxima: each term is at its maximum
- Number of roots: compare ranges of both sides or draw graphs

11. Summation of series

STEPS

1. Change every term to $\tan^{-1}$
2. Write $T_r = \tan^{-1}\frac{b-a}{1+ab} = \tan^{-1}b - \tan^{-1}a$ (split the denominator as $1+ab$)
3. Add: only first and last terms remain; let $n \to \infty$ for S_∞

Formulas

$$\sum_{r=1}^{n}\tan^{-1}\frac{1}{1+r(r+1)} = \tan^{-1}(n+1) - \frac{\pi}{4}; \quad \sum_{r=1}^{n}\tan^{-1}\frac{2r}{r^4+r^2+2} = \tan^{-1}(n^2+n+1) - \frac{\pi}{4}$$

$$\sum_{r=1}^{n}\tan^{-1}\frac{2}{r^2} = \tan^{-1}(n+1) + \tan^{-1}n - \frac{\pi}{4}, \quad S_\infty = \frac{3\pi}{4}$$

12. Common mistakes checklist

STEPS

1. Did I keep the answer inside the principal range?
2. Did I use $\sin^{-1}(\sin x) = x$ only when $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$?
3. Did I check the condition $xy < 1$ / $xy > 1$ in the $\tan^{-1}$ addition?
4. Did I use $\pi -$ for negative arguments of $\cos^{-1}$, $\cot^{-1}$, $\sec^{-1}$?
5. Did I reverse the inequality for $\cos^{-1}$, $\cot^{-1}$?
6. Did I substitute every root back into the original equation?